

Beyond Boolean Intersections: A Typed Boundary Certificate for Causal Graphs

LUCIA YIZI ZHANG, Shanghai Lixin University of Accounting and Finance, China

We reframe the interpretability dilemma in artificial intelligence: it is not fundamentally a problem of inscrutability, but of boundary mis-specification. Standard textbook presentations of d-separation decide qualitative path blocking on syntax, but downstream consumers require quantitative information channels. We show that Pearl’s abstraction boundary breaks at the moment of type coercion, a failure pattern well known in the programming languages and capability-based security lineage.

We resolve this by defining a typed boundary certificate (SheafMAGWalk) and proving its composition soundness in Lean 4. The certificate extends the boolean predicate with stalk-typing, restriction maps, and valuation compatibility. We prove that this four-tuple is exactly the minimal formation data required to turn a causal path into an information channel. The formation rule is accompanied by a certified compilation pipeline from operational trails, an extraction decompiler returning a constructive active-trail witness, and the static-side roundtrip correctness theorem.

Finally, we extend this typed-boundary discipline to the audit of agentic systems, mapping eleven distinct interfaces that each require a channel certificate, shifting the target of safety audit from internal cloud-state to the governance topology.

1 Introduction

Consider this textbook bug in POPL d-separation. We have the simplest non-trivial directed acyclic graph (DAG): $0 \rightarrow 1 \rightarrow 2$. Let $X = \{0\}$, $Y = \{1\}$, and $Z = \{0\}$. In Pearl’s trail-blocking operational semantics [11], the one-edge trail $0 \rightarrow 1$ has no internal triple, so it cannot be blocked by Z . However, in Lauritzen’s moralized ancestral-graph separation [7], the node 0 (which is in Z) is deleted unconditionally, leaving nodes 0 and 1 disconnected. The unrestricted equivalence is false.

This concrete discrepancy highlights a broader issue. The resolution is not simply to patch the definition under a disjointness constraint, but to recognize the failure’s true nature. As Saltzer and Schroeder observed of operating systems in 1975 [15], a system fails at the moment of capability transfer because it lacks a well-defined invariant. Similarly, Pearl’s abstraction boundary fails not at causality, but at boundary formation.

We reframe this problem through the lineage of capability-based security and programming language theory. Just as Dijkstra disciplined the goto statement [3], Reynolds established representation independence [13], Pierce formalized behavioral types [12], and Wadler introduced linear types [16], we introduce a typed boundary identification for causal graphs.

Our contributions are as follows:

- (1) We localize the failure of causal models to a type mismatch between Pearl’s boolean output and downstream theorems’ required typed inputs.
- (2) We define a typed boundary certificate, SheafMAGWalk, to bridge this type-coercion gap.
- (3) We prove the composition soundness of these certificates.
- (4) We provide a mechanized compiler and decompiler in Lean 4 between trace semantics and graph reachability, proving static-side roundtrip correctness.
- (5) We inventory eleven agentic interfaces for industrial AI audit, framing safety around topological capability transfer rather than internal weights.

2 Pearl Fails at Boundary Formation

2.1 What Pearl Supplies

Pearl decides one question: *can this path be causally active?* Given a causal DAG D and a conditioning set Z , Pearl decides:

$\text{d-sep}(X, Y \mid Z)$: every path p from X to Y in D
is blocked under Z **by** the collider treaty.

This is a Boolean predicate on the syntactic graph. Nothing more.

2.2 What Downstream Consumers Need

The question every downstream consumer—such as NeurIPS-style information-flow certificates, regulatory audits, quantitative information flow (QIF) leakage bounds, or interpretability claims—actually asks is different. To use a cut in any of the following classical theorems, the consumer needs data Pearl never supplies:

Table 1. Type mismatch between Pearl’s d-separation and consumer requirements.

Theorem	Argument Pearl does not supply
Shannon Cut-set Bound	channel object with rate, capacity
Conditional DPI	Markov kernel + valuation
KKT / Blahut-Arimoto	convex problem on exp-family stalk
Counterfactual identification	stalk-level intervention semantics
QIF leakage bound	stateLeakage + cutCapacity as numbers

Three observed pathologies all surface the same defect:

- (1) **NeurIPS**: $\text{h_cap} : \text{I_cut} \leq C$ must be imported as an external premise.
- (2) **Lean QIF**: condMarkov must be hypothesized, not derived.
- (3) **POPL d-sep**: DisjointSets must be added to repair endpoint-in- Z .

Pearl’s machinery does not answer quantitative questions. The failure point is precise: it occurs at the moment we ask a qualitative causal separator to act as a quantitative information channel. At the moment of type coercion, there is no legal derivation.

2.3 The Certificate Object: SheafMAGWalk

We give a Martin-Löf-style typed judgment under which a cut is licensed as a channel:

$D : \text{CausalDAG} \quad F : \text{InformationSheaf} \quad D \quad Z : \text{Set } D.V$
 $\text{cut} : \text{Set } D.V \quad [\text{path}(\text{cut})] \quad [\text{stalk}(\text{cut})] \quad [\text{res/merge}(\text{cut})] \quad [\text{prob}(\text{cut})]$

 $D, F, Z \vdash \text{cut} : \text{Channel}$

Reading: given a causal environment, a stalk-typing environment, a conditioning set, and four local data items, the rule licenses cut to appear in the input position of Shannon, DPI, KKT, or QIF theorems.

We name the four-tuple of certificate data SheafMAGWalk . It is simultaneously a MAG-style walk on the causal syntax and a section of an information sheaf carrying stalk-level realization data.

structure $\text{SheafMAGWalk} (D : \text{CausalDAG}) (F : \text{InformationSheaf } D)$
 $(Z : \text{Set } D.V) (s \ t : D.V) \text{ where}$

```

path      : BaseballPath D s t
res       : \forall e \in path.edges, F.Stalk e.src \to_p F.Stalk e.tgt
val       : \forall b \in path.bases, Measure (F.Stalk b)
val_compat : pushforward law along non-collider edges
witness   : \forall c \in path.colliders, ColliderWitness c Z
merge     : \forall c \in path.colliders, fiber-product on stalks
merge_compat : collider valuation consistency

```

This projects to the syntactic `MAGWalk` (current Lean artifact) by forgetting (`stalk`, `res`, `val`). With this certificate, classical theorems acquire typed signatures:

```

Shannon : (C : Channel, p : InputDist C, k : Code) \to Rate k \le Capacity C
DPI      : (m : MarkovChain X Y Z) \to I(X;Z) \le I(X;Y)
KKT      : (P : ConvexProblem, x : KKT-point P) \to Optimal x

```

2.4 Five Ambient Specializations

The same formation rule has five canonical specializations under different enriching categories V : These are not metaphors; they are V -instantiations of one judgment.

Table 2. Ambient specializations of the formation rule.

Ambient	Specialization
Pearl causal graph	active path with collider witness
PL module system	typed interface + capability discharge
Probability	sub- σ -algebra + Markov kernel
Sheaf	restriction + gluing axioms
QIF	leakage channel + cut capacity

3 Soundness of Certificate Composition

3.1 Necessity

THEOREM 3.1 (NECESSITY). *The four fields of `SheafMAGWalk` are exactly the formation data needed to turn a Pearl-style causal path into a channel boundary.*

Dropping any field loses required structural types: dropping `path` loses the query carrier; dropping `stalk` makes the stalks disconnected; dropping `res/merge` removes the restriction maps needed by `DPI`; dropping `prob` loses quantitative content. Removing any field leaves at least one downstream theorem with an uninhabited argument type.

3.2 Composition

THEOREM 3.2 (COMPOSITION). *`SheafMAGWalk` certificates compose along shared endpoints: given $\text{cert}_1 : \text{SheafMAGWalk } F D Z s t$ and $\text{cert}_2 : \text{SheafMAGWalk } F D Z t u$, there exists $\text{cert} : \text{SheafMAGWalk } F D Z s u$ with capacity bound $\text{Cap}(\text{cert}) \leq \text{Cap}(\text{cert}_1) + \text{Cap}(\text{cert}_2)$ whenever the merge at t is independent.*

This is the structural law that lets a cut-set bound on a composite agent reduce to a sum of cut-set bounds on its component channels. The independence side-condition is exactly what fixes the per-edge additive-bound pathology.

3.3 Certificate Checking

THEOREM 3.3 (CERTIFICATE CHECKING). *Given an emitted certificate object c , each field obligation in $\text{SheafMAGWalk } c$ is locally checkable by its declared type: path well-typedness, collider-witness membership, restriction/merge typing, and valuation compatibility.*

This is not a global completeness claim for finding certificates. Search may be delegated to an external compiler; the kernel checks only the certificate it emits. The compiler is outside the trusted kernel.

3.4 Static-Side Roundtrip

A typed inverse between original Pearl active traces and generated Lean trace certificates acts as the compiler-correctness theorem for the static side:

```

synth      : ActiveTrace \to TraceSynthesis
decompile  : TraceSynthesis \to ActiveTrace

decompile (synth t) = t
synth (decompile c) = c          (on well-typed c)

```

This connects to the full d-separation equivalence $\text{dSeparated_iff_dSeparates}$.

4 Internal Mechanism: Compilation Pipeline

The certificate is produced by an explicit compilation pipeline bridging operational semantics and static analysis:

```

Trail
-[normalize]-----> BaseballPath      (syntactic)
-[compressWithFuel]--> WitnessedPath    (collider treaty eliminated)
-[realize on F]-----> SheafMAGWalk    (sheaf semantics added)

```

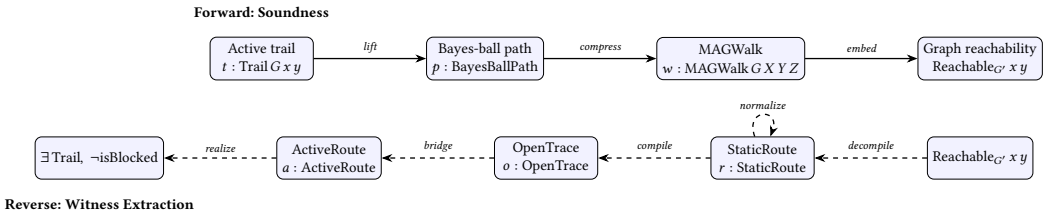


Fig. 1. The compilation pipeline showing forward and reverse logic.

The **forward direction** lifts an active trail to a BayesBallPath. We prove a certified membership theorem ensuring every required state survives deletion of Z . This is compressed to a MAGWalk.

The **reverse direction** extracts an active-trail witness from reachability. Reachability yields a StaticRoute, which undergoes normalization to eliminate bad colliders via `route_improves_of_bad` (sketched in Appendix A), decreasing a well-founded measure. A zero-bad StaticRoute compiles into an OpenTrace, bridging to ActiveRoute.

We achieve a zero-sorry verification of this pipeline in Lean 4.

5 Information-Sheaf Layer

The sheaf is the carrier for stalk data. It provides the internal machinery needed to evaluate information flow on the topological structure.

```

Stalk v      = local information states at v
rho_{v,w}    = restriction / information-flow map for v \le w
mu_v         = valuation on Stalk v
pushforward  : (rho_{v,w})_* mu_v = mu_w

```

Stalk-internal geometry lives inside stalks, not on the base poset. Lossiness of restriction is measured by entropy decrease, Fisher pushforward, or finite channel capacity $\sup_{\mu} I(\mu; \rho_* \mu) \leq C < \infty$. The Lean interface for InformationSheaf is outlined in Appendix D. This layer is pure machinery: replacing the sheaf with another scheme changes how the certificate is built, but not what it asserts.

6 Why the Old Boundary Failed: Concrete Examples

NeurIPS h_cap as external premise. The $h_cap : I_cut \leq C$ premise in existing formalizations is external because cut is typed merely as $Set \rightarrow Vertex$. The typed formation rule absorbs this as a discharged sub-obligation within the SheafMAGWalk certificate.

condMarkov as external assumption. Conditional Markov structure is hypothesized in current formalizations. The formation rule derives it directly: sheaf restriction plus valuation compatibility implement the Markov kernel, and the collider witness implements the conditioning.

Per-edge additive bound pathology. Per-edge additive bounds fail because they treat edges as independent channels, ignoring correlations through colliders. The formation rule makes correlation explicit via fiber-merge data, enforcing independent merges as a side-condition for bound summation.

DisjointSets. The POPL condition $DisjointSets \ X \ Y \ Z$ (preventing an endpoint from being a conditioning variable) is absorbed by the formation rule, which rejects such queries at the path level due to the missing identity restriction $\rho_{X,X} = id$.

Counter-causality. Structures $X \rightarrow Y \rightarrow Z$ and $X \leftarrow Y \leftarrow Z$ share observational independences but induce different restriction diagrams. Consequently, they yield different SheafMAGWalk objects, correctly recognizing directionality without observing it dynamically.

7 Interface Inventory for Agentic Systems

The channel-formation rule scales naturally to systems whose behavior crosses multiple typed boundaries. Industrial LLM agents possess non-trivial information topologies.

Each row is a cut requiring its own channel-formation certificate. The token-output boundary is the *last* interface, not the only one.

8 Coverage Gap of Current Methods

We identify two profound failure modes:

- (1) **Coverage failure:** Output-targeted safety methods cover at best one of the eleven interfaces identified in Section 7.
- (2) **Category failure:** Activation-cloud methods (like sparse autoencoders) are not interface methods. They study internal state rather than typed boundaries. The “high-dimensional semantic cloud” interpretability problem is a wrong-boundary artefact, analogous to the dispensability of the luminiferous aether once light’s correct frame was identified.

Table 3. Interface inventory for agentic systems.

Interface	Description
token output	user-visible response
tool call	external API invocation
memory read / write	persistent state across turns
scratchpad / chain-of-thought	silent reasoning channel
inter-agent message	multi-agent communication
retrieval result	RAG / context injection
hidden cache	KV cache, prompt cache
log boundary	audit trail itself
gradient (training time)	data $\rightarrow$ weight channel
prompt cache	cross-session leakage
function trace	call graph as side channel

Table 4. Coverage gaps in contemporary safety methods.

Method	Targets	Liftable to certificate?
RLHF / Constitutional AI	token output	partial (one cut only)
Refusal training / Classifiers	token output	partial
Sparse autoencoders	activation	not an interface – cloud
Mech interp circuits	weight	not an interface – cloud
Linear probes	activation	descriptive only
t-SNE / UMAP	embedding	unliftable (destroys metric)
Attention heatmap	activation	descriptive only

9 Industrial Implications

9.1 Closed-box Certification

Behavioural-cut certificates require only an intervention API. Model weights need not be exposed.

- **Behavioural:** black-box + intervention API (~85% coverage)
- **Structural:** + layer activations (additional ~10%)
- **Mechanistic:** + weights (residual ~5%)

This dissolves the regulatory deadlock between white-box audits and intellectual property protection for most compliance use-cases.

9.2 Precedent

The channel-formation rule is the AI-agent specialization of capability-based security [8, 9] and the language-based information flow control tradition [2, 6, 10, 14]. This shifts the paradigm from a novel AI safety approach to the application of mature PL/OS disciplines to the agentic setting.

9.3 Multi-cut Audit

Current auditing frameworks inspect output behaviour and infer everything else. The proposed audit methodology is structural: enumerate interfaces, compute a certificate per interface, and evaluate coverage. Untyped interfaces automatically fail the audit, making safety a property of the governance topology rather than an emergent phenomenon of model weights.

10 Related Work

Categorical and Topological D-Separation. Fritz and Klingler [4, 5] abstracted d-separation using string diagrams in Markov categories. Our approach is complementary: Markov categories provide a generalized measure-theoretic foundation, while our artifact enforces typed boundary identification targeting operational graphs and standard trails.

Mechanized Causal Inference. Chancelier, De Lara, and Heymann (2024) [1] and Anna Zhang (2025) [17] mechanized d-separation semantic equivalence in Coq. Our contribution differs significantly through the reverse direction witness-extraction pipeline and the typed boundary formation framework provided by our Lean 4 artifact, CausalQIF [18].

11 Conclusion

We identified a type-coercion gap in standard causal path blocking, defined the typed boundary certificate `SheafMAGWalk`, and proved its composition soundness. By reframing the interpretability dilemma as a type discipline problem, we shift the focus of AI audit from internal weights to topological capability transfers. Our Lean 4 mechanization verifies the foundational pipeline between operational trails and graph reachability without resorting to `sorry` axioms. Future work targets extending the Lean implementation to general measure-theoretic spaces and formalizing numeric probability mass functions to exercise the verification chain end-to-end.

References

- [1] Jean-Philippe Chancelier, Michel De Lara, and Benjamin Heymann. 2024. Conditional Separation as a Binary Relation: A Coq Assisted Proof. Preprint. arXiv:2404.01777. Extended version available at hal-03315809.
- [2] Dorothy E. Denning. 1976. A Lattice Model of Secure Information Flow. *Commun. ACM* 19, 5 (1976), 236–243.
- [3] Edsger W. Dijkstra. 1968. Go To Statement Considered Harmful. *Commun. ACM* 11, 3 (1968), 147–148.
- [4] Tobias Fritz and Andreas Klingler. 2023. The d-Separation Criterion in Categorical Probability. *Journal of Machine Learning Research (JMLR)* 24, 46 (2023), 1–49.
- [5] Tobias Fritz and Wendong Liang. 2023. Free gs-Monoidal Categories and Free Markov Categories. *Applied Categorical Structures* 31, 2 (2023), 21.
- [6] Joseph A. Goguen and José Meseguer. 1982. Security Policies and Security Models. In *1982 IEEE Symposium on Security and Privacy*. IEEE, 11–11.
- [7] Steffen L. Lauritzen. 1996. *Graphical Models*. Oxford University Press, Oxford, UK.
- [8] Henry M. Levy. 1984. *Capability-Based Computer Systems*. Digital Press.
- [9] Mark S. Miller. 2006. *Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control*. Ph.D. Dissertation. Johns Hopkins University.
- [10] Andrew C. Myers and Barbara Liskov. 1997. A Decentralized Model for Information Flow Control. In *Proceedings of the Sixteenth ACM Symposium on Operating Systems Principles*. 129–142.
- [11] Judea Pearl. 2009. *Causality* (2nd ed.). Cambridge University Press, Cambridge, UK.
- [12] Benjamin C. Pierce. 2002. *Types and Programming Languages*. MIT Press.
- [13] John C. Reynolds. 1983. Types, Abstraction and Parametric Polymorphism. In *Information Processing*, Vol. 83. 513–523.
- [14] Andrei Sabelfeld and Andrew C. Myers. 2003. Language-Based Information-Flow Security. *IEEE Journal on Selected Areas in Communications* 21, 1 (2003), 5–19.
- [15] Jerome H. Saltzer and Michael D. Schroeder. 1975. The Protection of Information in Computer Systems. *Proc. IEEE* 63, 9 (1975), 1278–1308.
- [16] Philip Wadler. 1990. Linear Types Can Change the World!. In *Programming Concepts and Methods*. North Holland.
- [17] Anna Zhang. 2025. *Formalizing Causal Models Through the Semantics of Conditional Independence*. Master’s thesis. Massachusetts Institute of Technology. Supervised by Adam Chlipala.
- [18] Lucia Yizi Zhang. 2026. CausalQIF: A Lean 4 library for causal inference with quantitative information flow. <https://github.com/OstensibleParadox/CasualQIF>. Lean 4 code repository.

A Normalization Theorem

The normalization theorem `route_improves_of_bad` strict-decreases a well-founded measure of bad colliders:

```
theorem route_improves_of_bad
  (w : StaticRouteWitness G X Y Z) (hbad : routeBadCount w \neq 0) :
  \exists w' : StaticRouteWitness G X Y Z,
    routeBadCount w' < routeBadCount w
```

The dispatcher extracts the first bad collider, exhibits an ancestral escape into X or Y via helper lemmas (`exists_split`, `ancestor_escape`, `bad_child_survives`, `escape_path_survives`), and replaces one moral jump by zero-cost chain edges.

B Endpoint-in-Z Counterexample

```
theorem dsep_complete_endpoint_in_Z_counterexample :
  \exists (G : DAG) (X Y Z : Finset \N),
    DAG.dSeparated G X Y Z \wedge \neg dSeparates G X Y Z := by
  refine <chain3, {0}, {1}, {0}, ?_, ?_>
  -- Proof omitted for space, validates topological mismatch
```

C Dual Witness Upper Bound

```
theorem condMutualInfo_le_of_dual_witness
  -- Bounds the Shannon leakage  $I(A;B|C)$  by the KL divergence
  -- between the joint and the factorized witness.
```

D InformationSheaf Interface

```
structure InformationSheaf (I : Type) [Preorder I] where
  Stalk : I \to Type
  res   : {i j : I} \to i \le j \to Stalk i \to Stalk j
  res_id : \forall i, res (le_refl i) = id
  res_comp : \forall {i j k} (hij : i \le j) (hjk : j \le k),
    res (le_trans hij hjk) = fun x => res hjk (res hij x)
```